RACE: Scalable Statistical Estimation of Functional Consistency in LLM Neurons

Runyu Wang[1], Bo Liu[2], Xiaxin Zhang[1], Yu Han[1], Jiawei Cao[1],
Xiaoye Zhang[3], Zhe Zhang[4], Yifan Yang[4], Peng Ping[1,*]

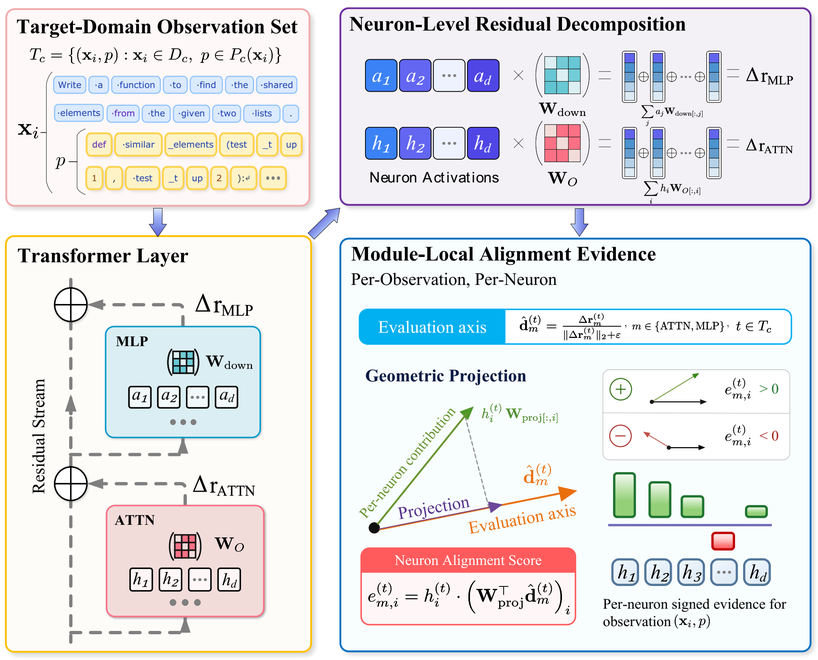
[1]School of Transportation and Civil Engineering, Nantong University
[2]Chongqing University of Post and Telecommunications [3]China Southern Power Grid Company Limited [4]Meituan

Stable residual alignment reveals neurons for **specific model behaviors.**

Forward-pass evidence ⟶ Bayesian consistency ⟶ Reference-set filtering

01 Collect Signed Evidence

Strong activations alone can hide inconsistent behavior. RDA collects signed residual evidence across all generated tokens (code/math) or teacher-forced tokens (reference/PyComp-1K).



Original paper, Fig. 1: residual-direction alignment (RDA). The axis is the current module's update, and each neuron supplies signed alignment evidence.

02 Estimate, Rank, and Filter

Functional consistency: positive mean evidence with low variability.

Forward-Pass Evidence

For neuron j at token-position observation t:

$$e_j^{(t)} = a_j^{(t)} \mathbf{w}_j^\top \hat{\mathbf{d}}^{(t)}, \qquad \hat{\mathbf{d}}^{(t)} = \frac{\Delta \mathbf{r}^{(t)}}{\|\Delta \mathbf{r}^{(t)}\|_2 + \varepsilon}.$$

Here $\mathbf{w}_j$ is the output-projection column; a_j is the MLP activation or attention output channel.

Bayesian Aggregation

Model $e_j^{(t)} \sim \mathcal{N}(\mu_j, \sigma_j^2)$ with a Normal-Inverse-Gamma prior. Update from the observation count, mean, and centered sum of squares.

$$\mu_j \mid \{e_j^{(t)}\} \sim t_{2\alpha_n}(\mu_{n,j}, \sigma_{\mu,j}), \qquad \sigma_{\mu,j}^2 = \frac{\beta_{n,j}}{\alpha_n \lambda_n}.$$

Conservative Alignment Magnitude (CAM):

$$\mathrm{CAM}_j = \mathbf{1}[\mu_{n,j} > 0] \max\{0, \mu_{n,j} - t_{.95,2\alpha_n} \sigma_{\mu,j}\}.$$

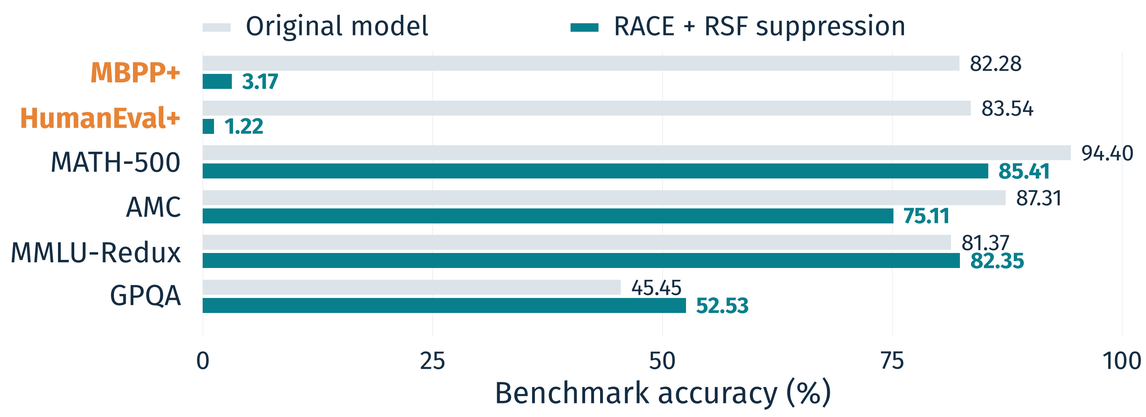
The lower credible bound penalizes noisy or scarce evidence. Default prior: $(\mu_0, \lambda_0, \alpha_0, \beta_0) = (0, 1, 1, 1)$; $\gamma = .05$.

Reference-Set Filtering (RSF)

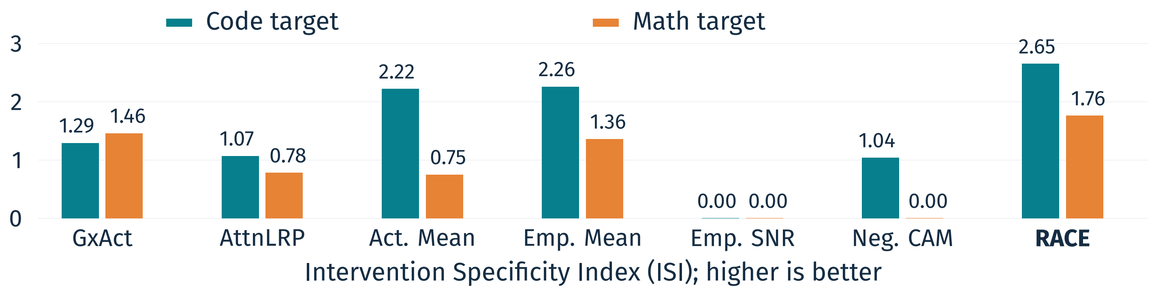
Remove neurons that rank highly on a general reference set (WikiText-2), then retain the top target neurons **within each layer and module**. Suppress their activations to test behavioral relevance.

03 Selective Capability Suppression

Code-selected MLP neurons: strong target and OOD drops, with much smaller effects on other benchmarks.



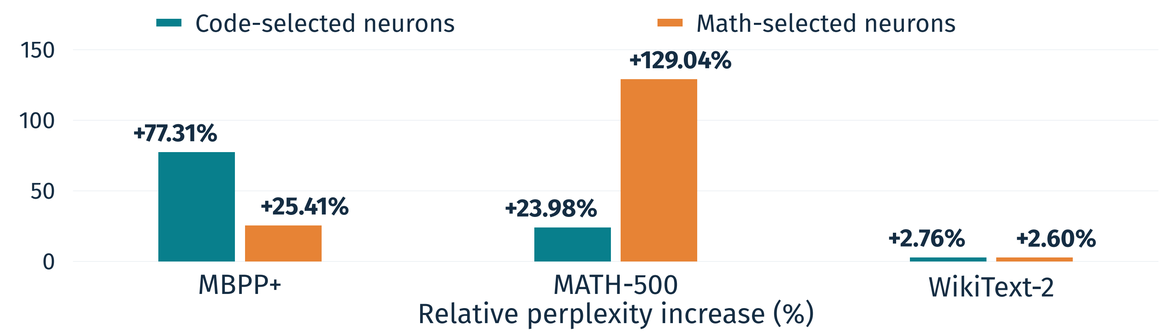
Qwen3-4B-it; MBPP+ scoring, WikiText-2 RSF, top-1% MLP suppression per layer. Accuracy averages three runs (Table 21); code targets are orange.



MLP interventions with RSF, same top-1% budget. ISI measures target drop relative to non-target drop; higher is better. Fig. 3; Tables 21 and 23.

04 Verify the Distributional Effect

Target-corpus perplexity rises sharply; the general reference corpus changes by less than 3%.



Qwen3-4B-it, RSF, top-1% MLP suppression per layer. Relative perplexity increase from Table 4; each color uses its own target scoring set.

The Effect Extends to 32B Models

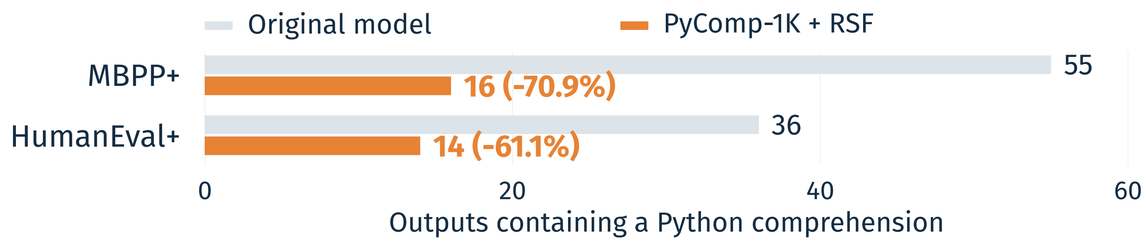
OLMo-3.1-32B-it, math scoring + WikiText-2 RSF; top-5% MLP suppression per layer (Table 3).

Benchmark	Original	Suppressed
MATH-500 (target)	86.40	**6.72**
AMC (OOD)	79.85	**7.46**
GPQA	48.60	37.79
MMLU-Redux	84.70	77.79

Scope: domain-specific effects are stronger in MLPs. Attention channels show greater cross-domain overlap, and RSF yields weaker attention specificity. Linear projections can miss nonlinear interactions.

05 Control a Narrow Code Behavior

PyComp-1K: 1,000 AST-verified Python statements targeting comprehensions. Suppress $K_{\mathrm{sel}} = 10$ neurons per layer after RSF.



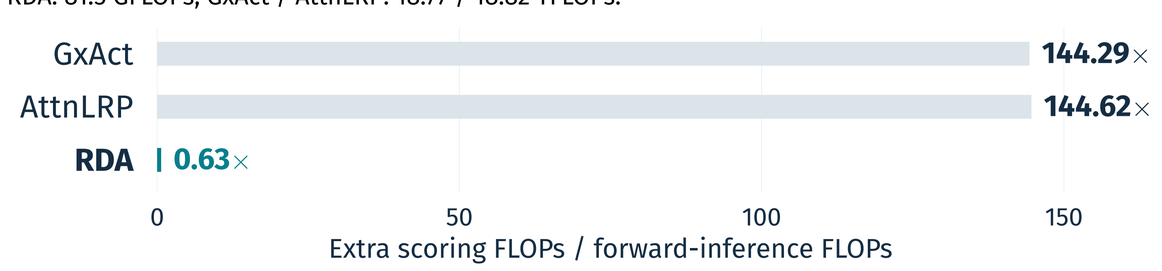
Qwen3-4B-it; number of outputs containing at least one comprehension (Table 5).

Functional correctness on the **remaining comprehension-using solutions**: MBPP+ 92.73% → 87.50%; HumanEval+ 86.11% → 85.71%. These are conditional subset scores.

06 Score at Forward-Pass Cost

≈ 230× lower extra scoring FLOPs

RDA: 81.5 GFLOPs; GxAct / AttnLRP: 18.77 / 18.82 TFLOPs.



Profiler on Qwen3-4B-it; 64 input / 16 output tokens (Tables 7 and 18). Extra scoring FLOPs exclude CAM/NIG updates, logging, and CPU aggregation; this ratio is not a wall-clock speedup.

Streaming: retain sufficient statistics per neuron; memory stays constant with corpus length. Full RACE on 378 MBPP+ prompts takes **587.28 s** on one RTX 4090 (Appendix T).

Paper: arXiv:2608.24758v2 **Code:** github.com/Nexround/RACE
Contact: 2430310032@stmail.ntu.edu.cn *Corresponding author: pingpeng@ntu.edu.cn